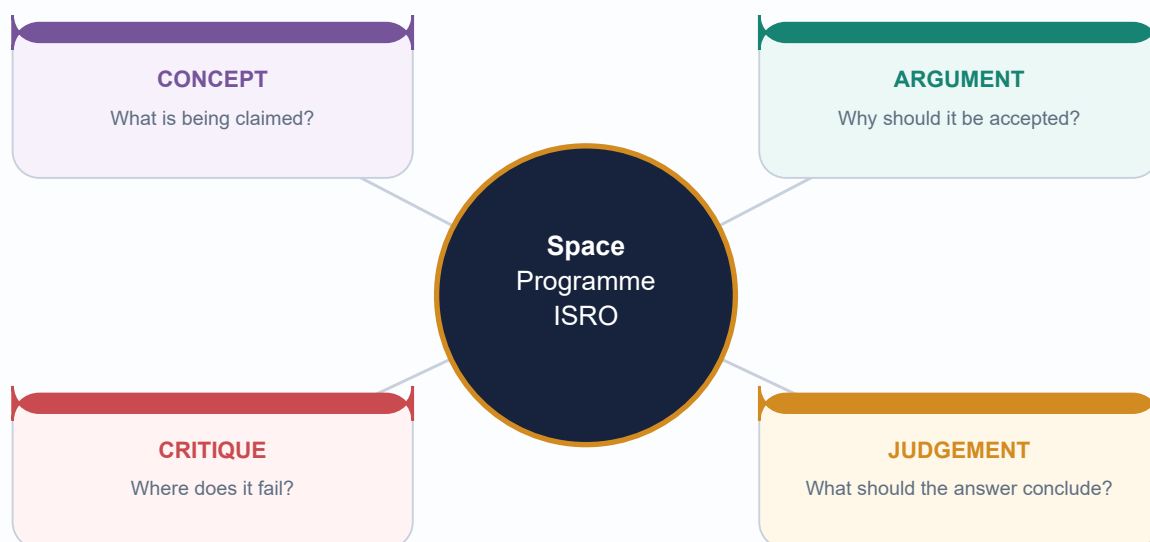


UPSC SCIENCE AND TECHNOLOGY WORKBOOK

Space Programme, ISRO, Launch Vehicles and Space Governance Solved-Practice Workbook

Subject: Science and Technology | Paper: GS-III
Eight source-trap MCQs with strict A -> B -> C -> D rotation
Original 10-, 15- and 20-mark solved content frameworks



Facts from official sources | Inferences clearly marked | No fabricated data

HIGH UPSC RELEVANCE

HIGH

Topic-Specific MCQ Loop 1

GS-III Science
and Technology

INTRO & ORIGIN

Each question is generated from an explicit source-chapter trap and explanation.

STATIC THEORY

- Q1. Which statement correctly resolves this common misconception: "ISRO, NSIL and IN-SPACE are just three names for the same organisation."?

- A. ISRO is the public agency; NSIL is the commercial PSU; IN-SPACE is the authorisation/promotional interface for private activity.

- B. ISRO, NSIL and IN-SPACE are just three names for the same organisation.

- C. PSLV, GSLV and LVM3 differ only by payload size.

- D. SSLV is simply a renamed PSLV variant.

- ANSWER: A - ISRO is the public agency; NSIL is the commercial PSU; IN-SPACE is the authorisation/promotional interface for private activity.

- Q2. Which statement correctly resolves this common misconception: "PSLV, GSLV and LVM3 differ only by payload size."?

- A. PSLV, GSLV and LVM3 differ only by payload size.

- B. Payload matters, but orbit profile, cryogenic capability and mission architecture also differentiate them.

- C. SSLV is simply a renamed PSLV variant.

- D. Space Policy 2023 means the government has withdrawn from the space sector.

- ANSWER: B - Payload matters, but orbit profile, cryogenic capability and mission architecture also differentiate them.

- Q3. Which statement correctly resolves this common misconception: "SSLV is simply a renamed PSLV variant."?

- A. SSLV is simply a renamed PSLV variant.

- B. Space Policy 2023 means the government has withdrawn from the space sector.

- C. SSLV is a separate small-satellite launcher concept with different market logic and infrastructure needs.

- D. NGLV or the Third Launch Pad should be described as already operational.

- ANSWER: C - SSLV is a separate small-satellite launcher concept with different market logic and infrastructure needs.

- Q4. Which statement correctly resolves this common misconception: "Space Policy 2023 means the government has withdrawn from the space sector."?

- A. Space Policy 2023 means the government has withdrawn from the space sector.

- B. NGLV or the Third Launch Pad should be described as already operational.

- C. "GSLV puts satellites into geostationary orbit."

- D. The policy separates public, regulatory and commercial roles; it does not abolish state leadership.

- ANSWER: D - The policy separates public, regulatory and commercial roles; it does not abolish state leadership.

MEMORY HOOK

Resolve the misconception before memorising the fact.

MAINS ANGLE**STUDY LINK**

Space Programme, ISRO, Launch Vehicles and Space Governance -> Source-grounded practice

HIGH**Topic-Specific MCQ Loop 2****GS-III** Science and Technology**INTRO & ORIGIN**

Each question is generated from an explicit source-chapter trap and explanation.

STATIC THEORY

- Q5. Which statement correctly resolves this common misconception: "NGLV or the Third Launch Pad should be described as already operational."?

- A. They are approved/future-capacity items, not current operational launch vehicles.

- B. NGLV or the Third Launch Pad should be described as already operational.

- C. "GSLV puts satellites into geostationary orbit."

- D. "Cryogenic and semi-cryogenic are the same thing."

- ANSWER: A - They are approved/future-capacity items, not current operational launch vehicles.

- Q6. Which statement correctly resolves this common misconception: "'GSLV puts satellites into geostationary orbit.'?"

- A. "GSLV puts satellites into geostationary orbit."

- B. It injects them into **geosynchronous transfer orbit**; the satellite's own propulsion completes the circularisation.

- C. "Cryogenic and semi-cryogenic are the same thing."

- D. "Indian Space Policy 2023 is India's space law."

- ANSWER: B - It injects them into **geosynchronous transfer orbit**; the satellite's own propulsion completes the circularisation.

- Q7. Which statement correctly resolves this common misconception: "'Cryogenic and semi-cryogenic are the same thing.'?"

- A. "Cryogenic and semi-cryogenic are the same thing."

- B. "Indian Space Policy 2023 is India's space law."

- C. Cryogenic = LOX + liquid **hydrogen** (flying on GSLV/LVM3 upper stages); semi-cryogenic = LOX + **kerosene**, still under development in India.

- D. ISRO, NSIL and IN-SPACe are just three names for the same organisation.

- ANSWER: C - Cryogenic = LOX + liquid **hydrogen** (flying on GSLV/LVM3 upper stages); semi-cryogenic = LOX + **kerosene**, still under development in India.

- Q8. Which statement correctly resolves this common misconception: "'Indian Space Policy 2023 is India's space law.'?"

- A. "Indian Space Policy 2023 is India's space law."

- B. ISRO, NSIL and IN-SPACe are just three names for the same organisation.

- C. PSLV, GSLV and LVM3 differ only by payload size.

- D. It is executive policy. India's binding obligations flow from the Outer Space Treaty family; there is no comprehensive domestic space statute yet.

- ANSWER: D - It is executive policy. India's binding obligations flow from the Outer Space Treaty family; there is no comprehensive domestic space statute yet.

MEMORY HOOK

Resolve the misconception before memorising the fact.

MAINS ANGLE**STUDY LINK**

Space Programme, ISRO, Launch Vehicles and Space Governance -> Source-grounded practice

HIGH**Solved 10-Mark Practice**

GS-III Science
and Technology

INTRO & ORIGIN

Explain Space Programme, ISRO, Launch Vehicles and Space Governance with reference to its concepts, applications and limitations.

ANSWER ARCHITECTURE**1. Concept**

Define the exact scope and terminology.

V

2. Core analysis

Organise the source content into two or three dimensions.

V

3. Evidence

Use the chapter's India-centric examples or chronology.

V

4. Critical layer

Add traps, limitations, counter-view or implementation constraints.

V

5. Judgement

Conclude in direct response to the directive.

STATIC THEORY

- ☐ **ISRO**: develops launch vehicles, spacecraft, national applications and flagship missions through centres such as VSSC, LPSC/IPRC and SDSC-SHAR.
- ☐ **SDSC-SHAR**: ISRO's spaceport and launcher-integration complex; the Launchers page describes it as responsible for launcher integration and operations.
- ☐ **IN-SPACe**: autonomous single-window nodal agency under DoS for enabling, authorising and supervising non-government space activities.
- ☐ **NSIL**: commercial PSU handling launch services, satellite-building services, satellite-based services and technology transfer/commercial production models.
- ☐ **Indian Space Policy 2023**: officially delineates roles of DoS, ISRO, IN-SPACe and NSIL rather than treating them as interchangeable entities.
- ☐ **Future-capacity projects referenced officially**: NGLV, Third Launch Pad and the Kulasekarapattinam spaceport belong to the next transport-infrastructure layer.
- ☐ PSLV has served Earth observation, navigation and other multi-payload missions; its exam value lies in versatility, not only in the word "polar."
- ☐ GSLV enabled launch of heavier communication payloads through an indigenous cryogenic upper stage.
- ☐ LVM3 is the heavy-lift platform for high-mass missions and the base launcher chosen for the human-spaceflight stack after human rating.
- ☐ NSIL creates commercial pathways for Indian industry participation in launcher production and launch services without replacing ISRO's public-science role.
- ☐ ☐ Reform has widened room for Indian private firms in manufacturing, launch support, downstream analytics and services, but the state still anchors safety and strategic capability.
- ☐ ☐ Limitation: launch cadence depends on testing, range safety, stage manufacturing, propellant logistics and integration capacity, not only on rocket design.
- ☐ ☐ Limitation: new projects such as SSLV scale-up, NGLV and additional pads demand long gestation, high capital expenditure and sustained industrial quality control.
- ☐ ☐ Limitation: India competes in a price-sensitive global launch market where reuse, scale and customer confidence matter as much as technical success.
- ☐ SLV-3 was India's first experimental satellite launch vehicle; ASLV served as an intermediate technology-learning step before PSLV.

MAINS ANGLE

These are model content points; arrange them into paragraphs rather than reproducing a list.

STUDY LINK

Space Programme, ISRO, Launch Vehicles and Space Governance -> 10-mark answer

HIGH

Solved 15-Mark PracticeGS-III Science
and Technology**INTRO & ORIGIN**

Analyse Space Programme, ISRO, Launch Vehicles and Space Governance with reference to its concepts, applications and limitations.

ANSWER ARCHITECTURE**1. Concept**

Define the exact scope and terminology.

V

2. Core analysis

Organise the source content into two or three dimensions.

V

3. Evidence

Use the chapter's India-centric examples or chronology.

V

4. Critical layer

Add traps, limitations, counter-view or implementation constraints.

V

5. Judgement

Conclude in direct response to the directive.

STATIC THEORY

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- ☐ **IN-SPACe**: autonomous single-window nodal agency under DoS for enabling, authorising and supervising non-government space activities.
- ☐ **NSIL**: commercial PSU handling launch services, satellite-building services, satellite-based services and technology transfer/commercial production models.
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- ☐ **Future-capacity projects referenced officially**: NGLV, Third Launch Pad and the Kulasekarapattinam spaceport belong to the next transport-infrastructure layer.
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- ☐ Reform has widened room for Indian private firms in manufacturing, launch support, downstream analytics and services, but the state still anchors safety and strategic capability.
- ☐ Limitation: launch cadence depends on testing, range safety, stage manufacturing, propellant logistics and integration capacity, not only on rocket design.
- ☐ Limitation: new projects such as SSLV scale-up, NGLV and additional pads demand long gestation, high capital expenditure and sustained industrial quality control.
- ☐ Limitation: India competes in a price-sensitive global launch market where reuse, scale and customer confidence matter as much as technical success.
- ☐ SLV-3 was India's first experimental satellite launch vehicle; ASLV served as an intermediate technology-learning step before PSLV.

MAINS ANGLE

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STUDY LINK

Space Programme, ISRO, Launch Vehicles and Space Governance -> 15-mark answer

HIGH

Solved 20-Mark PracticeGS-III Science
and Technology**INTRO & ORIGIN**

Critically examine Space Programme, ISRO, Launch Vehicles and Space Governance with reference to its concepts, applications and limitations.

ANSWER ARCHITECTURE**1. Concept**

Define the exact scope and terminology.

V

2. Core analysis

Organise the source content into two or three dimensions.

V

3. Evidence

Use the chapter's India-centric examples or chronology.

V

4. Critical layer

Add traps, limitations, counter-view or implementation constraints.

V

5. Judgement

Conclude in direct response to the directive.

STATIC THEORY

- ☐ **ISRO**: develops launch vehicles, spacecraft, national applications and flagship missions through centres such as VSSC, LPSC/IPRC and SDSC-SHAR.
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- ☐ ☐ Limitation: India competes in a price-sensitive global launch market where reuse, scale and customer confidence matter as much as technical success.
- ☐ SLV-3 was India's first experimental satellite launch vehicle; ASLV served as an intermediate technology-learning step before PSLV.

MAINS ANGLE

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STUDY LINK

Space Programme, ISRO, Launch Vehicles and Space Governance -> 20-mark answer

HIGH

Workbook Rapid Revision

GS-III Science
and Technology

INTRO & ORIGIN

Final revision is placed after teaching and solved practice.

TOPIC RECALL MAP

Space Programme, ISRO, Launch Vehicles and Space Governance

V

Definition ☐ ISRO: develops launch vehicles, spacecraft, national applications and flagship missions through centres such as VSSC, LPSC/IPRC and SDSC-SHAR.	Mechanism ☐ SDSC-SHAR: ISRO's spaceport and launcher-integration complex; the Launchers page describes it as responsible for launcher integration and operations.
Application ☐ IN-SPACE: autonomous single-window nodal agency under DoS for enabling, authorising and supervising non-government space activities.	Evaluation ☐ NSIL: commercial PSU handling launch services, satellite-building services, satellite-based services and technology transfer/commercial production models.

STATIC THEORY

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MEMORY HOOK

Meaning -> mechanism -> evidence -> limitation -> judgement

MUST-KNOW FACTS

1. □□ Limitation: new projects such as SSLV scale-up, NGLV and additional pads demand long gestation, high capital expenditure and sustained industrial quality control.
2. □□ Limitation: India competes in a price-sensitive global launch market where reuse, scale and customer confidence matter as much as technical success.
3. □ SLV-3 was India's first experimental satellite launch vehicle; ASLV served as an intermediate technology-learning step before PSLV.
4. □ ISRO's Launchers page lists PSLV, GSLV and LVM3 as active operational launch vehicles.
5. □ PSLV is described by ISRO as the workhorse launcher with multiple variants and a strong success record.
6. □ GSLV's indigenous cryogenic upper stage is the key distinction that enabled heavier communication payload launches.

UPSC TRAPS

WRONG: ISRO, NSIL and IN-SPACe are just three names for the same organisation.

CORRECT: ISRO is the public agency; NSIL is the commercial PSU; IN-SPACe is the authorisation/promotional interface for private activity.

WRONG: PSLV, GSLV and LVM3 differ only by payload size.

CORRECT: Payload matters, but orbit profile, cryogenic capability and mission architecture also differentiate them.

WRONG: SSLV is simply a renamed PSLV variant.

CORRECT: SSLV is a separate small-satellite launcher concept with different market logic and infrastructure needs.

WRONG: Space Policy 2023 means the government has withdrawn from the space sector.

CORRECT: The policy separates public, regulatory and commercial roles; it does not abolish state leadership.

MAINS ANGLE

Use Space Programme, ISRO, Launch Vehicles and Space Governance as an argument, not as an unstructured list of facts.

STUDY LINK

Science and Technology -> Space Programme, ISRO, Launch Vehicles and Space Governance

End of Register Notes - UPSC Agent / Copilot CLI